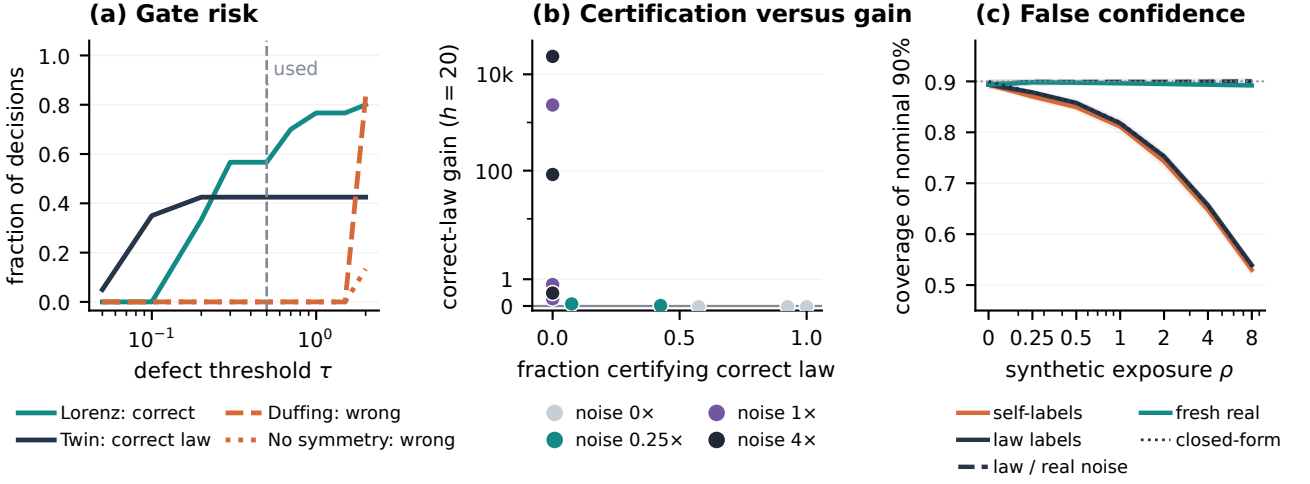
(a) Gate risk
fraction of decisions
1.0
0.8
0.6
0.4
0.2
0.0
used
10^{-1}
10^{0}
defect threshold τ
Lorenz: correct
Duffing: wrong
Twin: correct law
No symmetry: wrong
(b) Certification versus gain
correct-law gain ($h = 20$)
10k
100
1
0
0.0
0.5
1.0
fraction certifying correct law
noise 0×
noise 1×
noise 0.25×
noise 4×
(c) False confidence
coverage of nominal 90%
0.9
0.8
0.7
0.6
0.5
0
0.25
0.5
1
2
4
8
synthetic exposure ρ
self-labels
fresh real
law labels
closed-form
law / real noise